

EXPERIMENT 5

PLAYFAIR CIPHER

AIM

To implement the Playfair Cipher substitution technique using C.

ALGORITHM

1. Read the plaintext and keyword.
2. Construct the 5×5 key matrix.
3. Split the plaintext into pairs.
4. Apply Playfair encryption rules.
5. Display the encrypted text.

CODE

```
#include <stdio.h>
#include <string.h>

char matrix[5][5]={
    {'K','E','Y','W','O'},
    {'R','D','A','B','C'},
    {'F','G','H','T','L'},
    {'M','N','P','Q','S'},
    {'T','U','V','X','Z'}
};

void find(char ch,int *r,int *c){
    if(ch=='J') ch='I';
    for(int i=0;i<5;i++)
        for(int j=0;j<5;j++)
            if(matrix[i][j]==ch){
                *r=i;
                *c=j;
            }
}

int main(){
    char text[20];
    int r1,c1,r2,c2;

    printf("Enter Two Letters: ");
    scanf("%s",text);
```

```

find(text[0],&r1,&c1);
find(text[1],&r2,&c2);

if(r1==r2){
    printf("Encrypted: %c%c",
        matrix[r1][(c1+1)%5],
        matrix[r2][(c2+1)%5]);
}
else if(c1==c2){
    printf("Encrypted: %c%c",
        matrix[(r1+1)%5][c1],
        matrix[(r2+1)%5][c2]);
}
else{
    printf("Encrypted: %c%c",
        matrix[r1][c2],
        matrix[r2][c1]);
}

return 0;
}

```

OUTPUT

Output

Clear

Enter Two Letters: hl
hl
Encrypted: EE

=== Code Execution Successful ===

RESULT

Thus the Playfair Cipher substitution technique was implemented successfully using C.